

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	28 September 2026
Team ID	
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

The architecture shows the major ServiceNow application components, the user interface, workflow and approval logic, request data storage, notifications, and the teams involved in approval and fulfillment.

Architecture Diagram – Automated Network Request Management

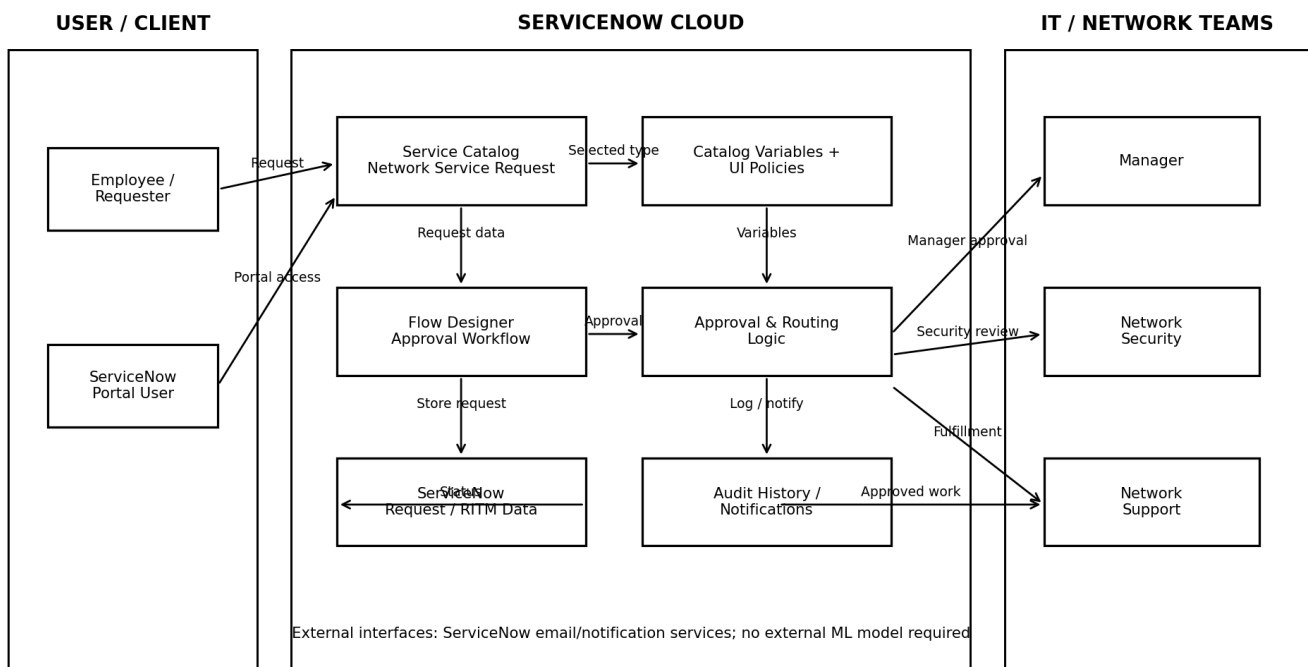


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Employee interface for submitting and tracking network service requests.	ServiceNow Service Catalog / Employee portal
2	Application Logic-1	Catalog form logic and dynamic fields based on request type.	Catalog Variables + Catalog UI Policies
3	Application Logic-2	Approval and routing workflow for manager and security approvals.	ServiceNow Flow Designer / Workflow Studio
4	Application Logic-3	Request assignment and fulfillment process for network teams.	Flow Designer actions, assignment groups and task workflow
5	Database	Stores catalog requests, requested items, variables, approvals and task information.	ServiceNow platform database / Glide records
6	Cloud Database	Platform-managed cloud data storage for ServiceNow records.	ServiceNow cloud platform
7	File Storage	No dedicated file-storage service is required for the core solution.	ServiceNow attachments / platform storage
8	External API-1	Email and notification delivery for request and approval updates.	ServiceNow Notification / Email services
9	External API-2	No external third-party API is required for the core implementation.	Not applicable
10	Machine Learning Model	No machine learning model is required for the current workflow-based solution.	Not applicable
11	Infrastructure (Server / Cloud)	Application runs on the ServiceNow cloud platform; development and testing use a ServiceNow developer instance.	ServiceNow Cloud / Developer Instance

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source application framework is required; the solution is built on ServiceNow platform capabilities.	ServiceNow platform / Flow Designer
2	Security	Role-based access, controlled approvals, request visibility, platform authentication, and audit history are used to protect network-service requests.	ServiceNow ACLs / Roles / Approval controls / Audit history
3	Scalable Architecture	The solution uses modular catalog, workflow, approval, notification and fulfillment components that can support additional network services.	ServiceNow platform, Flow Designer, reusable catalog components
4	Availability	The solution is hosted on the ServiceNow cloud platform and is accessed through the ServiceNow instance.	ServiceNow Cloud
5	Performance	Dynamic fields reduce unnecessary input, while automated routing and workflow processing reduce manual handoffs.	Catalog UI Policies / Flow Designer / ServiceNow platform